EMBC Manufacturability Roadmap

Idea-only, not tested or validated

Context: Uses the EMBC design context and models:

$$YCore = (n \cdot \mathrm{C_{12}H_{14}O_5Si(F_x)} + 2 \cdot \mathrm{XeN} + m \cdot (\mathrm{SiR_2O})] + [\mathrm{BNNT}] + [\mathrm{GaN}]$$

]), target resonance

$$f_0 = 13.56 \text{ MHz}$$

and latency constraint

$$t_p \leq 0.82 \text{ ns}$$

. This document is a conceptual, industry-oriented manufacturability roadmap and does **not** include procedural fabrication steps.

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1 High-level manufacturing strategy and staged pilot plan

Goal: Move from lab-scale concept to a pilot demonstrator that validates the EMBC material resonance and autonomous feedback metrics while minimizing exotic material usage and industrial risk.

Staged approach (conceptual):

- **Stage A — Lab Characterization (0–6 months)**
  - Validate phase properties: measure (\tilde{\varepsilon}(\omega)), (\sigma(\omega)) for YCore precursors, BNNT samples, and GaN films near

    $$f_0$$

    .
  - EM simulation of representative BNNT bundle microstructures to estimate

    $$L_{\mathrm{mat}}$$

    and mutual coupling.
  - Small coupon tests to validate EMT predictions and resonance behavior.
- **Stage B — Pilot Module Development (6–12 months)**
  - Produce modular test modules (mm-scale) using selective placement of BNNT inserts and GaN patches on silicate substrates.
  - Integrate metrology and test fixtures for S-parameter, resonance, and latency testing.
  - Iterate material formulations (YCore stoichiometry) to tune

    $$\varepsilon_{\mathrm{eff}}$$

    and loss.
- **Stage C — Foundry Partnership & Pilot Fabrication (12–18 months)**
  - Engage GaN-on-Si foundry for small pilot runs (die-level GaN integration).
  - Develop wafer-level compatible BNNT handling strategy (discrete inserts → patterned embedding).
  - Produce a small pilot batch (tens of modules) for system-level testing.
- **Stage D — Scale Readiness (post-18 months)**
  - Evaluate yield, cost per module, and regulatory/safety readiness.
  - Decide on scale path: foundry scale-up, hybrid contract manufacturing, or continued R&D.

Key stage objectives: Demonstrate resonance at

$$f_0$$

with measurable feedback magnitude (|G(\omega_0)|\ge1), verify latency budget

$$t_p \le 0.82\ \mathrm{ns}$$

for representative paths, and show repeatable module fabrication at pilot volumes.

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2 Required capabilities and equipment categories

High-level capability groups (conceptual):

- **Materials synthesis & handling**
  - BNNT sourcing and dispersion capability (wet/dry handling, alignment tools).
  - Silicate/YCore formulation (sol-gel, spin/coat, curing ovens).
  - GaN epitaxy and GaN-on-Si die/bonding access (foundry or epi house).
- **Deposition & patterning**
  - Thin-film deposition (PECVD, ALD, sputter) for GaN and silicate layers.
  - Patterning tools (photolithography, laser ablation) for selective GaN placement.
- **Bonding & integration**
  - Die/wafer bonding (hybrid bonding, adhesive bonding) for GaN-on-Si integration.
  - Precision placement tools for discrete BNNT inserts (micro-dispensing, pick-and-place for inserts).

- **Metrology & test**
  - Vector Network Analyzer (VNA) with probes for S-parameter and resonance measurement near 13.56 MHz.
  - Time-domain reflectometry (TDR) or high-speed oscilloscopes for latency measurement (sub-ns resolution).
  - SEM/TEM for microstructure imaging; AFM for surface topology; X-ray CT for 3D inspection.
  - Dielectric spectroscopy equipment for (\tilde{\varepsilon}(\omega)) and (\sigma(\omega)).
- **EM simulation & modeling**
  - Full-wave EM solvers (FEM/FDTD) for microstructure inductance and field distributions.
  - Multiphysics tools for thermal/CTE analysis.

Conceptual equipment list (categories only): deposition tools, bonding stations, micro-placement tools, VNA + probes, high-speed oscilloscope/TDR, SEM/TEM, dielectric spectrometer, EM simulation licenses.

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3 Modular integration approaches to minimize exotic material use

Principles: Reduce exotic material volume, isolate high-risk materials to small areas, and use modular testable units.

Approaches (conceptual):

- **Selective placement (preferred early stage)**
  - Place BNNT bundles and GaN patches only where needed (resonant cells), not across full wafer. This reduces BNNT consumption and simplifies processing.
- **Discrete inserts / hybrid modules**
  - Build modules where BNNT bundles are prepared as discrete inserts (micro-pellets or patterned islands) that are embedded into the silicate host at die level.
- **GaN die bonding (die-level integration)**
  - Use small GaN die or epitaxial islands bonded to Si substrates rather than full GaN wafer processing to reduce cost and leverage foundry services.
- **Hybrid bonding & interposers**
  - Use interposer layers for routing and to decouple thermal/CTE mismatches between GaN and silicate host.
- **Testable tile architecture**
  - Design small tiles (mm²) that can be tested individually and tiled later; this isolates failures and reduces scrap cost.

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4 Process windows and compatibility constraints (conceptual)

Key compatibility constraints (conceptual):

- **Temperature budgets**
  - GaN epi and GaN-on-Si processes may tolerate high temperatures during epitaxy but bonded GaN die and BNNT may limit post-processing temperatures. Define a maximum process temperature

    $$T_{\max}$$

    (pilot assumption; to be measured) and ensure silicate curing and any anneals are compatible.

- **Coefficient of thermal expansion (CTE)**
  - Mismatch between GaN, BNNT-composite, and silicate host can induce stress. Use CTE analysis to bound allowable thermal cycles. Conceptual constraint:

    $$|\alpha_{\text{GaN}} - \alpha_{\text{host}}| \leq \Delta\alpha_{\max}$$

    where (\Delta\alpha_{\max}) is a manufacturability tolerance to be defined.
- **Chemical compatibility**
  - Solvents, etchants, and adhesion promoters used for silicate/YCore must not degrade BNNT or GaN surfaces. Define a compatibility matrix for candidate chemistries.
- **Electrical and thermal conduction**
  - Ensure BNNT/GaN conductive paths do not create unintended shorts; include isolation layers and controlled vias.
- **Percolation threshold awareness**
  - Avoid uncontrolled BNNT percolation that would dramatically change (\sigma_{\text{eff}}). Treat percolation volume fraction (v_c) as a hard design parameter in process windows.

Design rule (conceptual): All process steps must be validated against a compatibility checklist that includes (T_{\max}), CTE mismatch, solvent exposure, and percolation risk.

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5 Supplier and partner types and selection criteria

Supplier / partner categories (conceptual):

- **GaN epi houses / foundries**
  - Criteria: GaN-on-Si capability, small pilot runs, die bonding services, process documentation, IP/NDAs.
- **BNNT suppliers / nanomaterials labs**
  - Criteria: material purity, batch consistency, supply volume, dispersion support, MSDS availability.
- **Materials formulators (silicate/YCore)**
  - Criteria: experience with sol-gel and low-temperature curing, ability to tailor dielectric properties, analytical support.
- **Metrology & test houses**
  - Criteria: VNA/TDR expertise, dielectric spectroscopy, EM measurement capability at 13.56 MHz.
- **EM simulation and integration partners**
  - Criteria: experience with anisotropic composites, microstructure inductance modeling, and multiphysics coupling.

Engagement suggestions (conceptual):

- Start with small, technical partnerships (university labs, specialty suppliers) for BNNT and YCore formulation.
- Use foundries with GaN pilot programs for die-level integration.
- Select partners who provide traceable material data and are willing to collaborate on iterative pilot runs.

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6 Pilot BOM and conservative cost estimate per module (conceptual)

Pilot module definition: a single test tile ~1 mm² area (conceptual) with selective BNNT inserts and GaN patch, packaged for test.

Conceptual pilot BOM (per module, illustrative):

- Silicate/YCore precursor (trace amount) — negligible material cost per tile.
- BNNT (micro-bundle) — high-cost specialty material; conceptual cost contribution significant at low volumes.
- GaN die/patch (small) — moderate cost (foundry/epi).
- Substrate (Si wafer piece) — commodity.
- Processing (deposition, bonding, patterning) — foundry/contract processing cost.
- Test & metrology (per-module amortized) — VNA/TDR time, SEM imaging.

Conservative per-module cost (conceptual):

- **Materials:** $50–$500 (BNNT cost dominates at low volume).
- **Processing (pilot foundry/contract):** $200–$2,000.
- **Test & metrology (per module amortized):** $50–$500.
- **Total conservative R&D module cost: $300–$3,000** per module (very approximate; depends on BNNT sourcing and foundry access).

Note: Costs scale down with volume and process maturity; BNNT supply and integration method are the largest cost drivers.

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7 Quality, test and measurement plan and required metrology tools

Key metrics to validate (conceptual):

- **Resonance frequency:** measure peak at

$$f_0 = 13.56 \text{ MHz}$$

  and tolerance (e.g., ±1%).
- **Feedback magnitude:** measure closed-loop gain (|G(\omega_0)|).
- **Latency:** measure propagation time (t_p) across representative paths; verify

$$t_p \leq 0.82 \text{ ns}$$

  .
- **Q-factor and losses:** extract (Q) from resonance bandwidth and compute (\sigma_{\text{eff}}).
- **Self-healing metrics (conceptual):** measure recovery rate after induced perturbation (non-destructive electrical/thermal perturbation) and quantify regeneration time constant.
- **Material properties:** (\tilde{\varepsilon}(\omega)), (\sigma(\omega)), BNNT orientation/distribution.

Required metrology tools (conceptual):

- **Vector Network Analyzer (VNA)** with low-frequency capability around 13.56 MHz and appropriate probes.
- **High-speed oscilloscope / TDR** for sub-ns latency measurement.
- **Dielectric spectrometer** for complex permittivity.
- **SEM/TEM, X-ray CT** for microstructure imaging and BNNT distribution.
- **Thermal cycling chamber** for CTE and thermal stability tests.
- **EM simulation tools** for correlating measured data with models.

Test plan outline (conceptual):

1. Material characterization (dielectric & conductivity).
2. Single-tile resonance and Q measurement.
3. Closed-loop feedback test and phase alignment checks.
4. Latency measurement across representative routing.

5. Robustness tests (thermal cycles, repeated perturbations) to observe self-healing metrics.

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8 Risk register and mitigation strategies (conceptual)

Risk	Impact	Likelihood (conceptual)	Mitigation (conceptual)
BNNT percolation → uncontrolled conductivity	High	Medium	Control volume fraction; use discrete inserts; measure percolation threshold (v_c).
BNNT anisotropy causing directional mismatch	High	High	Use anisotropic EMT modeling; align BNNT intentionally; design directional tests.
Inability to realize required L_{mat} from microstructure	High	Medium	EM simulation; iterate bundle geometry; consider hybrid inductive structures (GaN loops).
GaN integration thermal/CTE stress	High	Medium	Limit post-bonding thermal budget; use compliant interposers; select foundry processes with low thermal impact.
Low yield at pilot scale	High	Medium	Modular tile approach; early foundry engagement; statistical process control.
Safety/regulatory issues with BNNT or Xe compounds	High	Low-Medium	Obtain MSDS; EHS review; limit exposure; regulatory consultation.

Conceptual contingency: Maintain alternative material strategies (e.g., replace BNNT with other high-aspect conductive inclusions) if supply or safety issues arise.

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9 Timeline and milestones for 18-month R&D → pilot

Month 0–3 (Kickoff & materials): finalize assumptions table, secure BNNT samples, GaN die options, and silicate precursor formulations. Complete initial dielectric and conductivity measurements.

Month 3–6 (Modeling & lab validation): EM simulations for BNNT bundles; small coupon EMT validation; initial resonance tests.

Month 6–9 (Pilot module design): design modular tiles, select integration approach (selective placement or discrete inserts), and finalize test fixtures.

Month 9–12 (Pilot fabrication runs): produce first pilot tiles (dozens), perform S-parameter, latency, and self-healing tests; iterate materials.

Month 12–15 (Foundry pilot & scale readiness): engage foundry for small pilot run; validate process windows and yield; update BOM and cost model.

Month 15–18 (System validation & decision): system-level testing, regulatory readiness checks, and decision gate for scale vs further R&D.

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10 Deliverables and decision gates per stage

Stage A deliverables: measured (\tilde{\varepsilon}(\omega)), (\sigma(\omega)) for phases; EM simulation report; assumptions table updated.

Decision gate A → B: measured material data consistent with EMT predictions and ability to reach target

$$\varepsilon_{\text{eff}}$$

and acceptable loss (Q).

Stage B deliverables: functional modular tiles with measured resonance and latency; test reports; updated BOM.

Decision gate B → C: repeatable tile fabrication with acceptable per-tile performance and pilot cost estimate within budget.

Stage C deliverables: foundry pilot wafers/dies, pilot batch test results, yield estimates, regulatory checklist.

Decision gate C → D: pilot yields and costs support scale decision or require further R&D.

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11 Executive summary and one-page checklist

Executive summary (conceptual):
This roadmap outlines a staged, industry-compatible path to pilot the EMBC architecture. It emphasizes modular tiles, selective placement of BNNT and GaN, early EM simulation to validate

$$L_{\text{mat}}$$

targets, and a rigorous measurement program to verify resonance at

$$f_0 = 13.56\ \text{MHz}$$

and latency

$$t_p \leq 0.82\ \text{ns}$$

. The largest technical risks are BNNT anisotropy/percolation and achieving the required inductance from microstructure; these are mitigated by anisotropic EMT modeling, discrete inserts, and EM simulation-driven geometry design. Conservative per-module pilot costs are estimated at **$300–$3,000** (illustrative). The plan targets a pilot decision within 18 months.

One-page checklist (conceptual):

- [ ] Finalize Assumptions Table and tolerances.
- [ ] Secure BNNT samples and GaN die options.
- [ ] Complete EM microstructure simulations for

$$L_{\text{mat}}$$

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- [ ] Validate EMT on coupons and measure (\tilde{\varepsilon}(\omega)).
- [ ] Build and test modular tiles for resonance and latency.
- [ ] Run Monte-Carlo robustness analysis with measured distributions.
- [ ] Engage foundry for pilot run if pilot metrics pass decision gates.
- [ ] Complete safety/regulatory reviews and MSDS collection.

Notes and disclaimers: This roadmap is conceptual and intentionally avoids procedural fabrication steps. All cost figures, timelines, and parameter values are illustrative and must be replaced with measured data and supplier quotes during execution. For any physical prototyping, follow institutional safety and regulatory procedures.